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DEPARTMENT OF BIOMEDICAL ENGINEERING

MS PROJECT REPORT

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Master of Science in Neural Engineering

CHARACTERIZATION OF NANOPARTICLE INTERACTIONS IN PERIPHERAL NEURAL
LESION MODELS

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Project Abstract

Neural lesions induce the trans-differentiation of Schwann cells (SCs) to a repair phenotype, wherein repair SCs recruit and participate with macrophages to clear myelin debris in lesion areas (myelin clearance). Inadequate clearance results in failure to complete nerve repair. Nanoparticle-based therapies enhance drug delivery by acting as carriers that improve tissue specificity, bioavailability, and release kinetics. Certain NPs- particularly chondroitin sulfate NPs (CS-NPs)- have demonstrated immunomodulatory potential. Notably, nanoparticles are of comparable size to myelin debris, raising the possibility that SCs may internalize them during clearance. However, the impact of CS-NP dosing on SCs remains unclear.

This project will characterize CS-NP interactions with SCs to address fundamental questions regarding uptake dynamics and cytotoxicity. CS-NP internalization will be evaluated with confocal microscopy across multiple doses. CS-NP cytotoxicity will be assessed by quantifying metabolic activity in dosed SCs using MTS assays. Competitive uptake assays will gauge the effects of CS-NP dosing on simulated myelin clearance in macrophage models. Anticipated outcomes include defining dose-dependent uptake dynamics, clarifying competitive interactions with myelin debris during clearance, and determining whether nanoparticle treatment modulates myelin clearance. Furthermore, the project aims to provide insight into how NP therapies could be leveraged to improve neural repair.

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List of Symbols and Abbreviations

(In order of appearance)

Central Nervous System	CNS
Schwann Cell	SC
Peripheral Nervous System.....	PNS
Wallerian Degeneration	WD
Spinal Cord Injury	SCI
Reactive Oxygen Species	ROS
Peripheral Nerve Injury	PNI
Oligodendrocyte Progenitor Cell	OPC
Chondroitin Sulfate.....	CS
Nanoparticle	NP
Chondroitin-Sulfate Nanoparticle.....	CS-NP
Lipopolysaccharide.....	LPS
Alzheimer's Disease	AD
Human Schwann Cell	hSC
Immortalized Bone Marrow-Derived Macrophage	iBMDM
Poly-L-Lysine	PLL
Hank's Balanced Salt Solution	HBSS
Poly (lactic acid-co-glycolic acid)-polyethylene glycol	PLGA-PEG
Blank Nanoparticle	BNP
Medium Molecular Weight Chondroitin Sulfate Nanoparticle	MCS-NP
Desulfated Chondroitin Sulfate Nanoparticle	DCS-NP
Chondroitin Sulfate-A	CS-A
Poly (lactic acid-co-glycolic acid)-polyethylene glycol-Chondroitin Sulfate.....	PLGA-PEG-CS
Bicinchoninic Acid Assay	BCA
Bovine Serum Albumin	BSA
Methyl tetrazolium Salt	MTS
Blank Nanoparticle	BNP
4',6-diamidino-2-phenylindole.....	DAPI

Introduction

The Role of Myelin in the Nervous System

Myelin is a protein found abundantly coating the axons of neurons throughout the nervous system. The two myelinating glial cells populating the nervous system are oligodendrocytes in the Central Nervous System (CNS), and Schwann cells (SCs) in the Peripheral Nervous System (PNS). The myelin sheaths provide insulation for axons and are key to signal conduction in the nervous system. Myelin sheath degeneration is a symptom common to various nervous system disruptions, including many chronic neurodegenerative diseases, spinal cord injury, and neural trauma. Lingering myelin debris is neuroinflammatory and is an agent of signal conduction deficits in patients with neural disease.

On Neural Injuries and Myelin Clearance

Signal conduction in the nervous system occurs along the axon of neurons. The action potential- a propagating gradient of membrane depolarization- triggers neurotransmitter release at axon terminals. This release drives synaptic transmission, wherein neurotransmitters diffuse along the synaptic cleft and bind to receptors on the postsynaptic cell, which integrates the incoming signal and initiates membrane depolarization. The speed of signal conduction along an axon is dependent on the axon membrane's internal impedance and the axon's internal contents. Axons with higher membrane resistance and lower internal resistance have much faster action potential propagation than their counterparts. Myelin is present in the membranes of glial cells that facilitate rapid transmission of neural signals. Nerves use myelinating glial cells to form myelin sheaths by wrapping around axons. These sheaths are electrical insulators for axons and are one of the greatest

contributors to their conduction velocity. Neural lesions result in an impacted myelin sheath, leading to impaired neural cell communication due to disrupted impulse generation and conduction.[1]

Nerve injury leads to acute axon degeneration caused by primary injury and progressive degeneration of the axon distal to the lesion site- a process known as Wallerian degeneration (WD) (Figure 1). Furthermore, inflammatory responses in the lesion site inhibit myelin production and promote myelin degradation. The process of myelin clearance involves the internalization of the fragmented myelin sheath (myelin debris) by phagocytes and is critical to nerve regeneration.[1-3] In self-resolving injuries, myelin clearance is completed, remyelination is enhanced, and tissue remodeling and functional recovery are achieved.[4] However, non-resolving injuries, such as in spinal cord injuries (SCI), are distinctly characterized by the sustained presence of myelin debris.[5] Incomplete clearance of myelin debris interrupts axon regeneration, limits remyelination, hinders functional recovery, and is associated with poor therapeutic outcomes. Moreover, long-term deposited myelin debris is a known source of neuroinflammation, inducing reactive oxygen species (ROS) in macrophages and microglia. Axonal degeneration is characteristic of neurodegenerative, neuroinflammatory, and neurotraumatic diseases. This project focuses on applications to degenerative states in the PNS sequential to trauma such as in peripheral nerve injury (PNI).[2]

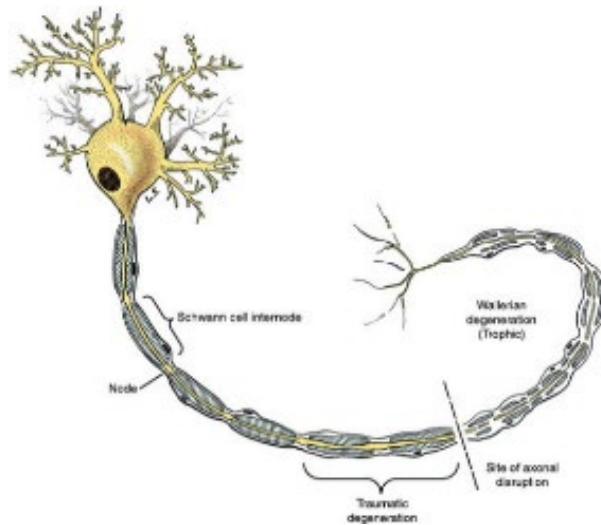


Figure 1: Wallerian degeneration of a neuron following lesion, PNS.[6]

The process of myelin clearance in PNI involves the trans-differentiation of SCs from a maintenance phenotype to a repair phenotype. The first week of myelin clearance is characterized by SC autophagy- a process by which SCs break down their own myelin and results in degradation of approximately 50% of their myelin sheaths. Macrophages recruited by the repair SCs infiltrate the distal segment of the lesion site after 2-3 days and shift to a pro-regenerative M2 polarization. The M2-polarized macrophages participate in myelin clearance through phagocytosis of deposited myelin debris, promoting neural injury repair.[7, 8]

Oligodendrocyte progenitor cells (OPCs) are recruited in the final stages of myelin clearance, as myelin presence is interruptive of OPC recruitment. Once recruited, OPCs proliferate and migrate towards lesion sites and differentiate into myelinating SCs. SCs attach to the lesioned axons and reinsulate axons, regenerating their myelin sheaths. This remyelination process is key to restoring native structure and function and is inhibited by sustained myelin presence. Delayed remyelination results in scarring that further inhibits functional recovery.[4]

Current Approaches to Resolving Non-clearing Myelin Deposition

Incomplete clearing of myelin deposition is common in non-self-resolving neural trauma such as SCI or PNI. PNIs have high prevalence in the U.S., adversely affecting both motor handling and sensory perception. Recovery time significantly depends on intervention time, with best outcomes seen in patients who seek treatment soon after injury. Moreover, axonal regeneration is a slow process (1-2 mm/day), and additional injury further delays recovery. PNI often results in lifelong disability. There are six common surgical approaches for PNI recovery: direct nerve repair, nerve grafting, nerve transfer, biomaterials, nerve conduits, and cell-based therapy. Non-surgical approaches include pain-mitigating medications (though not directly impacting functional recovery), electrical stimulation, and physical therapy.[9]

Current research aims to stimulate or enhance remyelination and prevent axon degeneration. Cell-based approaches include promoting endogenous remyelination of the myelin sheath through the stimulated differentiation and enhancement of OPCs, inhibiting integrin-mediated neuro-inflammation, and transplantation of myelinating glial cells.[4, 10] Moreover, chondroitin sulfate (CS) have been shown to support myelin clearance and may lead to functional recovery.[11]

Nanoparticles (NPs) are tunable biomaterials that enhance drug delivery by acting as carriers that improve tissue specificity, bioavailability, and release kinetics. Moreover, NPs have also demonstrated immunomodulatory and neuroprotective potential.[12] NP-integrated scaffolds are considered to have great remodeling potential, tackling common drug-delivery issues including trafficking and controlled delivery. Chondroitin sulfate nanoparticles (CS-NPs) are NPs that are surface functionalized or encapsulated with CS. CS is highly biocompatible, given its natural presence within the ECM, inherent biodegradability, low immunogenicity, and non-cytotoxic profile.[13]

Literature Review

CS-NPs and CS-derived-NPs are increasingly used as drug-delivery platforms. CS-chitosan NPs, for instance, have shown potential as nanocarriers for bromfenac sodium (nonsteroidal anti-inflammatory drug used to treat ocular pain and inflammation after cataract surgery), enhancing permeation by a factor of 1.62, and improving retention.[14] CS-NPs have also shown high targeting potential, demonstrating targeting of CD44 (over-expressed in tumors) for the treatment of breast cancer[15], agglomeration in the Golgi apparatus of cancer cells[16], targeting of hepatic stellate cells in fibrotic liver pathologies,[17] among others.

Immunomodulatory effects of CS-NPs have also been explored - in particular, in applications of chronic inflammatory diseases. CS-coated NPs, for example, have successfully scavenged ROS in induced osteoarthritis models. Moreover, NP treatment inhibited proinflammatory polarization of macrophages and their proinflammatory cytokines, suppressing osteoarthritis progression, protecting cartilage, and improving outcomes.[18] Similar results have been shown in rheumatoid arthritis models, where CS-modified tragacanth gum-gelatin composite nanocapsules loaded with curcumin nanocrystals showed significantly reduced expression of IL-1 β in lipopolysaccharide (LPS)-stimulated macrophages, suppressing joint inflammation and supporting joint tissue restoration.[19]

Applications of NPs in neurodegenerative diseases have also been explored, albeit to a lesser extent. For instance, Selenium-CS-NPs have shown neuroprotective properties in relation to Alzheimer's disease (AD) by inhibiting amyloid- β agglomeration, mitigating oxidative stress, and protecting cells from cytotoxic factors.[20] Moreover, other studies have hinted at neuroprotective outcomes, including mitigation of pro-inflammatory cytokines, reduced effect of ROS, and

inhibition of tau phosphorylation, translating to improved spatial learning, memory, and cognitive health in AD mice.[21]

CS has been used in conjunction with other nanoplateforms (particularly electrospun nanofibers) to modulate glial cell adhesion, migration, and proliferation to facilitate neural regeneration.[22] These glial responses are critical to guiding axonal regrowth following nerve injury, and support CS dose-dependent regeneration in nanotherapeutics. Moreover, these findings suggest that nanoscale presentation of CS can enhance SC activity.

Although CS-based NPs have been studied in several cancer models, chronic inflammatory conditions, and in several neural contexts including AD, interactions with SCs in neural lesion models remain poorly characterized. Existing studies have examined how CS in nanostructured scaffolds may enhance SC adhesion and migration, suggesting that CS nanomaterials could modulate SC behavior. However, there are no studies evaluating cellular uptake, cytotoxicity, or dose-dependent responses of CS-NPs in SCs.

This report aims to address knowledge gaps relating to NP interactions in neural lesions, including dose-dependent uptake dynamics, cytotoxicity, and competitive uptake interactions.

Methods

Effects of CS-NP treatment in SC cultures were evaluated in terms of direct uptake, cytotoxicity, and competitive uptake (with myelin debris) in vitro. Moreover, these effects were also studied in macrophage cultures, and in a macrophage-SC co-culture. Human Schwann cells (hSCs) were generously provided by the Dietrich laboratory, who isolated cells from donor tissue under

approved ethical guidelines. Immortalized Bone Marrow-Derived Macrophages (iBMDM) gifted by the Vaccari laboratory and maintained at our laboratory were used for experiments.

Poly (lactic acid-co-glycolic acid)-polyethylene glycol (PLGA-PEG) polymer NPs are used for all experiments. Cells were dosed with one of three NP formulations- blank nanoparticles (BNPs), medium-molecular weight chondroitin sulfate nanoparticles (MCS-NPs, 30 – 100 kDa), or desulfated chondroitin sulfate nanoparticles (DCS-NPs, used as a secondary control to account for sulfation). Fluorescent analogs of each NP formulation were prepared with CF647-amine (Biotium).

Statistical calculations performed by SAS JMP Student Edition 18. Graphs generated using GraphPad Prism 8.0.2.

Nanoparticle Fabrication Method

Three NP formulations are used in the experiments described in this report: BNPs, MCS-NPs, and DCS-NPs. These NPs were fabricated and fluorescently labeled in-house (CF647-amine) and stored at -20 C until use. NP characteristics (including size, zeta potential, and polydispersity index) are shown in Table 1.

Table 1: Experimental NPs: Characterization

NP	Statistic	PDI	Size (nm)	Zeta Potential
BNP	Mean	0.2285	129.2	-23.64
	95% CI	[0.05269, 0.4042]	[95.30, 163.0]	[-33.26, -14.03]
MCS-NP	Mean	0.1867	153.4	-20.70
	95% CI	[0.1487, 0.2246]	[128.8, 177.9]	[-30.60, -10.80]
DCS-NP	Mean	0.2548	93.98	-28.63
	95% CI	[0.004602, 0.5049]	[68.54, 119.4]	[-29.94, -27.31]

PLGA-PEG-CF647 Fabrication

To fluorescently label the PLGA-PEG polymer, 10 mg of PLGA-PEG-COOH was dissolved into DMSO. Fluorophore conjugation was initiated using a 1 mL aqueous EDC/NHS solution (15.2 mg EDC and 3.6 mg sulfo-NHS). This activation reaction is allowed to proceed for 15 minutes at room temperature with continuous shaking (1000 rpm).

After activation, 30 μ L of CF-647 amine was added, and the conjugation reaction was carried out for two hours on a thermomixer at room temperature. After fluorophore attachment, m-dPEG₁₂ amine was added to the solution as a capping reagent to quench any active functional groups and prevent unintended crosslinking. Conjugation proceeded for an additional hour under the same conditions.

Following conjugation, the reaction mixture was diluted 1:1 with Milli-Q water, then purified and fractionated by centrifugal ultrafiltration (3 kDa MWCO, 2000 x g, 45 minutes). The concentrate was reduced to a final volume of < 2 mL, frozen, and lyophilized to produce a dry, blue-tinted PLGA-PEG-CF647 powder. The purified polymer was resuspended in DMF at 50 mg/mL and stored at -20 C.

BNP Fabrication

BNPs were made by first combining a solution of 10 mg/mL PLGA-PEG polymer in DMF and 10 mg/mL PLGA-PEG-CF647 in DMF. The polymer mixture was added drop-wise (one drop every 4-5 seconds) 10 mL MilliQ water stirred at 800 rpm, inducing nanoprecipitation. The solvent was allowed to evaporate for two hours. BNPs were then washed twice by centrifugal ultrafiltration (100 kDa MWCO, 2000 x g, 20 minutes) and then resuspended in MilliQ water at 10 mg/mL. The BNPs were then lyophilized and stored at -20 C.

MCS-NP Fabrication

MCS-NP were made through the following steps: CS-A digestion, polymer fabrication and conjugation, and nanoprecipitation. CS-A was enzymatically depolymerized using Hyaluronidase Type II (7.2 U/mg CS-A) for 24 hours in 10x PBS, followed by overnight desalting in 100-500 Da dialysis tubing. The digested CS solution was fractionated using 3 kDa and 10 kDa MWCO filters to obtain low (<3 kDa), medium (3-10 kDa), and high (>10 kDa) fractions. The fractions were frozen and lyophilized to a dry powder, then resuspended at 50 mg/mL and stored at -20 C.

Digested CS-A (5x molar excess) was added to 10 mL of a 1 mg/mL PLGA-PEG-NH₂ solution in DMSO following activation with EDC (19 mg) and sulfo-NHS (10 mg) for 15 minutes. Conjugation proceeded for 6-7 hours at room temperature on a magnetic stir plate. The reaction was diluted with 10 mL Milli-Q water and dialyzed overnight (12-14 kDa MWCO). The polymer solution was concentrated via centrifugal ultrafiltration (10 kDa MWCO, 2000 x g, 30 minutes), frozen, and then lyophilized overnight to a dry PLGA-PEG-CS powder.

5 mg of PLGA-PEG-CS was dissolved into a MilliQ water/DMF mixture. Separately, 5 mg of PLGA-PEG-CF647 was dissolved in DMF at 10 mg/mL and combined with the CS-polymer solution. The resultant mixture was added drop-wise (one drop every 4-5 seconds) into 10 mL of MilliQ water stirred at 800 rpm. MCS-NPs were formed over the course of two hours while the solvent evaporated. MCS-NPs were washed twice by centrifugal ultrafiltration (100 kDa MWCO, 2000 x g, 20 minutes) and suspended in MilliQ water at 10 mg/mL. The MCS-NPs were then lyophilized and stored at -20 C.

DCS-NP Fabrication

DCS-NPs production follows a three-step approach: first, the digestion and desulfation of chondroitin sulfate, followed by polymer fabrication and conjugation, then finally nanoprecipitation. CS-A underwent hydrolytic acid depolymerization, where HCl acid was added to an aqueous CS-A solution until pH 0.5 was reached. The mixture was heated to 60 C with constant stirring for ten hours to induce chain shortening and desulfation. The solution was then neutralized with NaOH to a pH of 7. The DCS was dialyzed and desalted overnight using 100-500 Da dialysis tubing. The DCS solution was then fractionated into low, medium, and high fractions (<3 kDa, 3-10 kDa, > 10 kDa, respectively) by centrifugal ultrafiltration. Fractions were lyophilized to a dry, yellow-brown powder, then resuspended at 50 mg/mL and stored at -20 C.

Digested and desulfated CS-A (5x molar excess) was added to 10 mL of PLGA-PEG-NH₂ (1 mg/mL in DMSO) after activation with EDC (19 mg) and sulfo-NHS (10 mg) for 15 minutes. Polymer conjugation proceeded for 6-7 hours at room temperature while stirred. Following conjugation, 10 mL of milli-Q water was added, and the mixture was dialyzed overnight (12-14 kDa MWCO). The polymer was concentrated (10 kDa MWCO, 2000 x g, 30 minutes), frozen, then lyophilized to a dry powder.

5 mg PLGA-PEG-DCS was dissolved in a DMF/Milli-Q water mixture and combined with 5 mg PLGA-PEG-CF647 (10 mg/mL in DMF). The mixture was added dropwise (one drop every 4-5 seconds) into 10 mL of Milli-Q water stirred at 800 rpm, allowing NP formation during two hours of solvent evaporation. DCS-NPs were washed twice (100 kDa MWCO, 2000 x g, 20 minutes), resuspended at 10 mg/mL, lyophilized, and stored at -20 C.

Cell Culture Maintenance

SCs were maintained in tissue-culture treated 75 cm² (T-75) flasks. SC growth medium was prepared by supplementing DMEM with 10% FBS, 1% penicillin-streptomycin, neuregulin (10 ng/mL), forskolin (2 μM), and gentamicin (0.25 μg/mL). iBMDMs were maintained in tissue-culture treated 75 cm² (T-75 flasks) containing iBMDM growth medium (DMEM supplemented with 10% FBS and 1% penicillin-streptomycin).

All flasks used for hSC maintenance were coated with 5-6 ug/mL laminin solution. All plates used for hSC seeding were coated with 0.1 mg/mL poly-L-lysine (PLL) and 15-20 ug/mL laminin solution. Sterile Hank's balanced salt solution (HBSS) was used as a buffer and wash solution for all tissue and cell processing steps.

Myelin Debris Isolation

Myelin debris was isolated from dissected brain and spinal cord (extracted through hydraulic extrusion, as shown in figure 2) tissue of three mice provided post-mortem by the Islet Immunoengineering Lab under approved animal protocols. Brain and spinal cord were dissected, minced into ~1 mm segments, then enzymatically dissociated in liberase (0.5 mg/mL in HBSS) at 37 C for 30 minutes with intermittent inversion to prevent settling. Tissue suspensions were then triturated, centrifuged at 300 x g for 5 minutes, and resuspended in HBSS. Myelin was then isolated using density gradient centrifugation (725 x g, low acceleration, no braking, 15 minutes). Myelin debris (thin white band in suspension) was collected and washed three times at 10,000 x g for 10 minutes each before resuspension in HBSS. Myelin used in imaging experiments was fluorescently labeled with Fluoromyelin (1:150, $\lambda = 555$ nm).

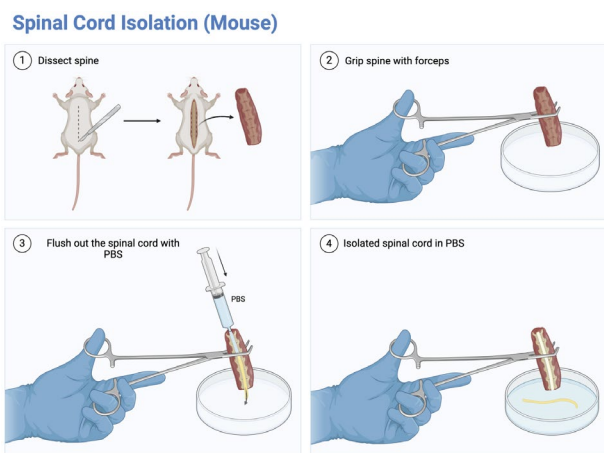


Figure 2: Spinal cord isolation using hydraulic extrusion.[23]

Myelin Debris Isolation and Characterization

The Pierce bicinchoninic acid (BCA) protein assay (Thermo Scientific) was used to assess the concentration of myelin protein in the isolated myelin debris using bovine serum albumin (BSA) as the standard. Absorbance was measured at $\lambda = 562$ nm using an Agilent BioTek Synergy H1 multimode reader.

A BCA assay was used to quantify isolated myelin debris. Linear interpolation determined the average sampled myelin concentration to be 0.975 mg/mL.

Cytotoxicity Assays

Cellular metabolism and viability in hSCs and iBMDMs following NP treatment was assessed using the methyl tetrazolium salt (MTS) colorimetric metabolic activity assay (Abcam). Absorbance was measured at $\lambda = 490$ nm using an Agilent BioTek Synergy H1 multimode reader. Cell viability was quantified against a standard curve following an eight-point two-fold serial dilution, starting at 20,000 cells. The MTS assay was performed for BNPs, MCS-NPs, or DCS-NPs for one- or three-day exposures. Cells were dosed at 1.2, 0.6, 0.3, or 0.15 mg/mL of NP solution for each MTS experiment.

Blank NP Direct Uptake Experiments

Fluorescent confocal microscopy was used to qualitatively examine BNP uptake in hSCs. Five-mm cover slips were sterilized and coated with PLL and laminin in 96-well plates. Each well received 100 μ L of 300,000 hSCs/mL suspension and incubated at 37 C for three days. HSCs were then dosed with 1.2, 0.6, 0.3, or 0.15 mg/mL fluorescent BNP solutions and incubated for two hours. After incubation, hSCs were stained with CellMask Green (1:1000) for 15 minutes, then fixed with 100 μ L of 4% PFA for 15 minutes. Cells were then counterstained with 4',6-diamidino-2-phenylindole (DAPI, 1:2000) for 2 minutes. Wells were washed, and discs were transferred and mounted to microscope slides for imaging.

Images were taken using a Nikon Eclipse TI inverted microscope using a 20x objective and excitation wavelengths of $\lambda = 488$ nm (CellMask Green), 647 nm (Alexa Fluor 647), and 461 nm (DAPI). Image processing was performed in FIJI/ImageJ. Confocal microscopy was sequentially performed to increase result confidence.

Live Cell Imaging, Competitive Uptake Experiments

Live cell imaging was conducted to quantify the impact of NP dosing on hSCs or iBMDMs that were simultaneously dosed with myelin debris. Wells precoated with PLL and laminin were seeded with 100 μ L of 200,000 hSCs/mL, which were allowed to incubate for 48 hours. After 24 hours, 100 μ L of 50,000 iBMDMs/mL were seeded in non-coated wells. The seeded iBMDMs were treated with 100 ng/mL LPS and incubated for 24 hours. The following day, iBMDM wells were treated with 5 ng/mL IFN- γ and incubated for two hours. Media was then changed on all wells using media containing 1:200 dilution CellBrite Green and allowed to incubate for 20 minutes at

37 C. Wells were washed three times with warm media, then dosed with MCS-NPs or DCS-NPs and a variable concentration of isolated myelin debris (0.6 mg/mL or 0.3 mg/mL).

Live cell imaging was also used to evaluate time-dependent hSC-uptake of MCS-NPs and DCS-NPs given a dose of 0.6 mg/mL or 0.3 mg/mL.

Images were taken using an Opera Pheonix Plus High Content Screening System (PerkinElmer) every hour for up to 12 hours using a 63x objective and excitation wavelengths of $\lambda = 488$ nm (CellBrite Green), 647 nm (Alexa Fluor 647), and 555 nm (Fluoromyelin).

Results

BNP-Treated hSCs: Uptake Experiment

HSCs treated with fluorescently labeled BNPs at 1.2, 0.6, 0.3, and 0.15 mg/mL showed no apparent uptake after a two-hour incubation period. Two images (hSCs with no BNP dosage, and a 10 mg/mL BNP sample) were used as qualitative controls. Sample images are shown in Figure 3. A nuclear membrane stain (DAPI, blue), and a membrane stain (CellMask Green, green) were used to identify cells.

Results suggest that there was no BNP internalization after two hours. HSCs dosed with 0.15 – 0.6 mg/mL BNP showed no evidence of internalization. However, hSCs dosed at 1.2 mg/mL showed BNPs extracellularly. Moreover, these BNPs appear agglomerated, likely due to the higher concentration (cells with lower doses showed no extracellular agglomeration).

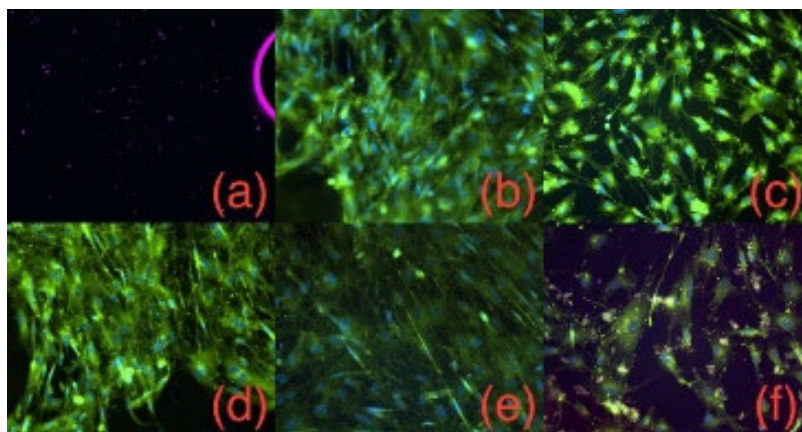


Figure 3: hSC BNP uptake experiment. Panel (a) shows the 10 mg/mL BNP sample. Panels (b-f) show hSCs dosed at 0 mg/mL (b), 0.15 mg/mL (c), 0.30 mg/mL (d), 0.6 mg/mL (e), and 1.2 mg/mL (f) BNP.

Images taken at 20x magnification.

NP-Treated hSCs: Cytotoxicity Assays

A cytotoxicity analysis of BNP treatment in hSCs was performed using an MTS assay. Six replicates were used for each dosage. Linear regression was conducted on a standard curve to predict sample concentration from measured absorbance values ($R^2 = 0.8568$, $p < 0.0001$).

A one-way ANOVA was used to evaluate if there were any significant differences between interpolated concentrations of treatment groups. There was no significant difference between the means of the treatment groups ($p = 0.6022$, $\alpha = 0.05$). Thus, hSCs treated with BNPs at 1.2, 0.6, 0.3, and 0.15 mg/mL show no significantly different viability when compared to an untreated control following a 24-hour incubation period (BNPs are not a significant effector of cell viability given a 24-hour incubation period, and within a dosing range of 0.15 to 1.2 mg/mL). Figure 4 below depicts a one-way analysis of cell count by treatment.

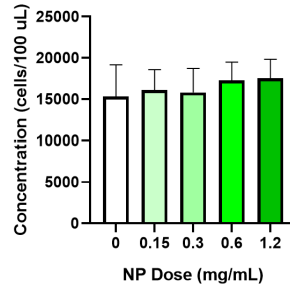


Figure 4: One-way analysis of hSC concentration (cells/100 µL) by BNP treatment. No significant difference was found between group means ($p = 0.6022$, $\alpha = 0.05$).

A separate analysis was conducted using MCS-NP and DCS-NP treatments with dosing equal dosing intervals to the BNP cytotoxicity assay and equal incubation period. Linear regression was performed on a standard curve, predicting concentration from absorbance values ($R^2 = 0.9498$, $p < 0.0001$). A one-way ANOVA was used to evaluate if there were any significant differences between interpolated concentrations of NP-treatment groups and an untreated control at the same seeding density. Results from the ANOVA ($p = 0.2670$) suggested that there were no differences between group means. Thus, both MCS-NP and DCS-NP dosing is not accompanied by significant changes in cell viability 24 hours post-treatment. Figure 5 below shows the group means and 95% confidence intervals.

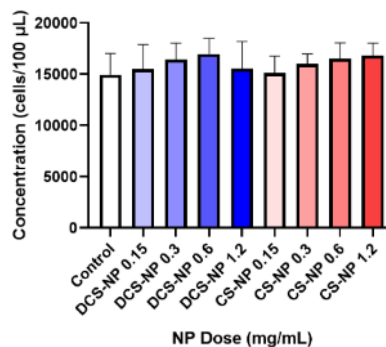


Figure 5: MTS Assay with MCS-NP, DCS-NP One-day Incubation. Sample Concentration (cells/100 µL) Means

To determine effects of NP incubation time on cytotoxicity, a contrasting three-day incubation dosing experiment was conducted, where hSCs were dosed at 1.2, 0.6, 0.3, or 0.15 mg/mL of MCS-NPs or DCS-NPs, with cells being treated at 24, 48, and 72 hours after seeding. Samples were treated with MTS reagent 96 hours post-seeding, and absorbance values were measured. Linear regression was conducted on absorbance values to interpolate sample concentrations ($R^2 = 0.8590$, $p < 0.0001$). A one-way ANOVA was used to evaluate significant differences between treatment groups. Group means and confidence intervals are shown in Figure 6.

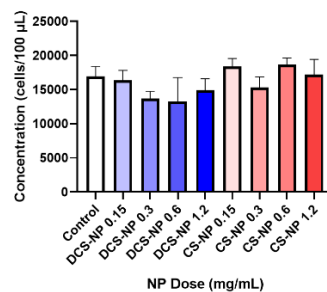


Figure 6: MTS Assay, 3-Day Extended Incubation with MCS-NP, DCS-NP Treatment. Sample Concentration (cells/100 µL) Means

The one-way ANOVA suggested there was at least one group with significantly different cell viability after three-day incubation ($p < 0.0001$). Post-hoc multiple comparisons were performed (contrasts) to evaluate between-group differences. A summary of conducted contrasts is shown in Figure 7.

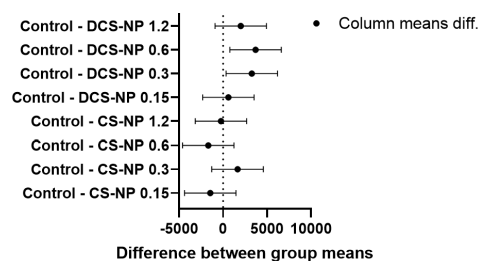
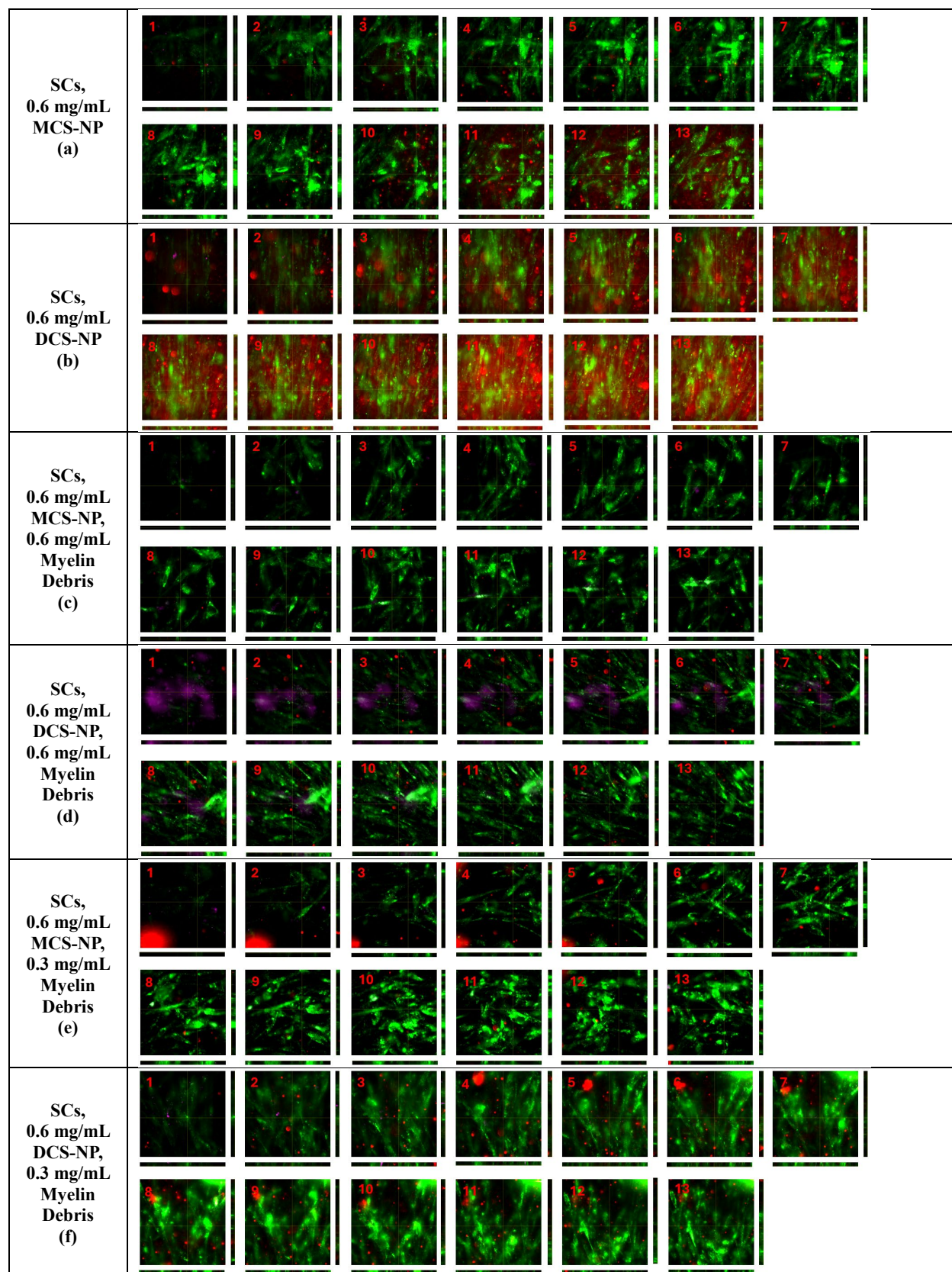


Figure 7: MTS Assay, 3 Day Extended Incubation Multiple Comparisons

Multiple comparisons suggest that there is a statistically significant difference between the control group and the DCS 0.6 mg/mL and DCS 0.3 mg/mL dosage groups. The DCS group, having a smaller mean concentration, may thus have cytotoxic potential.

Live Cell Imaging of Competitive Uptake Model

Live cell imaging of MCS-NP- and DCS-NP-treated hSCs and iBMDMs was conducted to establish what interactions, if any, occur in isolated myelin-internalizing cells. A 2x2x2 full factorial design was set-up, with iBMDMs and hSCs treated with myelin debris at a concentration of either 0.6 or 0.3 mg/mL, followed by treatment with either MCS-NPs or DCS-NPs at a fixed concentration of 0.6 mg/mL (duplicates were used for every condition). Two separate controls (iBMDM and hSCs, each with only myelin treatment) were used to contrast uptake dynamics. Representative images are shown in Figure 8.



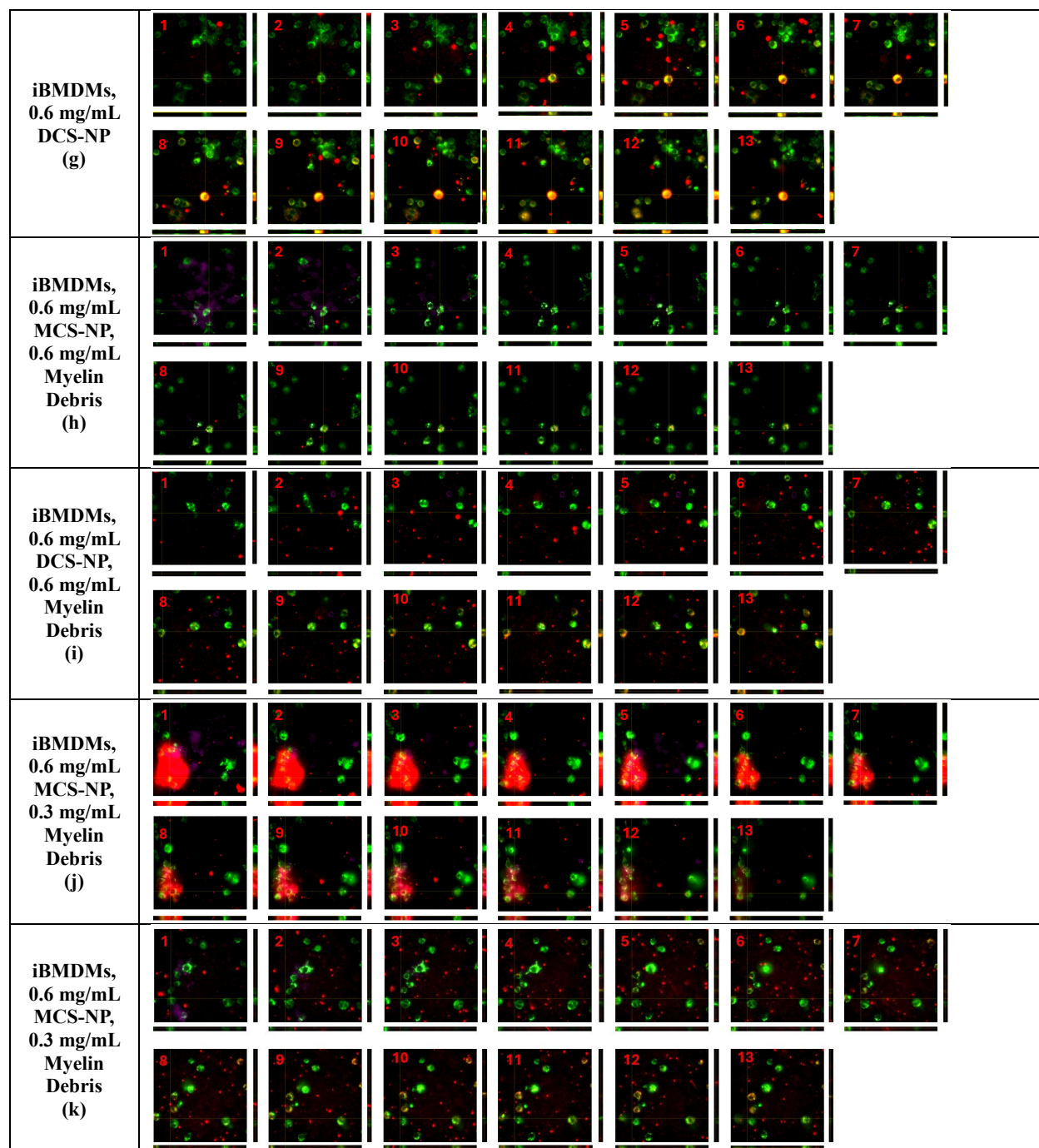


Figure 8: Competitive Uptake Assay, Live Cell Imaging with 12-hour incubation. Images taken every 1 hour from $t = 0$ (1) to $t = 12$ (13). Panels (a) to (f) show competitive uptake experiments with SCs, while panels (g) to (k) show iBMDM experiments (see panel for treatment description). Images taken at 63x.

Discussion and Conclusions

Analyses of hSCs following BNP uptake suggest there are no internalized NPs after a two-hour incubation period. Similar experiments conducted in the past with iBMDMs have suggested that cells may internalize NPs briefly after treatment but are promptly evicted into clumps. Image processing of the 1.2 mg/mL BNP-dosed cells suggests that a similar effect may occur with hSCs. Observable NPs are agglomerated and do not intersect with the stained membranes. Control images from competitive uptake experiments suggest that DCS-NPs and MCS-NPs may both accumulate around or within the cell membrane of the SCs. This differs from macrophages, which readily show internalization of both NPs by four hours post-treatment.

Cytotoxicity assays for BNPs, MCS-NPs, and DCS-NPs mostly suggest that there is no main effect of NP dosing on SC metabolism or viability. Moreover, the extended three-day incubation MTS assay provided evidence of there being no cytotoxic effects with prolonged exposure or with degraded MCS-NPs. These results coincide with MTS assays run on LPS-stimulated iBMDMs with the same NPs, suggesting that MCS-NP-dosing in neural lesion contexts (mostly populated by SCs and iBMDMs) are not cytotoxic. DCS-NPs demonstrated some cytotoxic potential in the extended incubation MTS, with 0.6 mg/mL and 0.3 mg/mL treatment groups significantly decreasing cell concentration.

Competitive uptake evaluations were used to qualitatively determine if co-treatments of NPs and myelin debris had any impact on NP uptake patterns by both cell types. SCs showed no visible differences in terms of NP membrane aggregation. Myelin debris signal also decreased across time in SC wells, with cells notably wrapping around the myelin. Across all treatments, SCs showed minimal NP accumulation around the membrane. Further experimentation is required to

determine if this is truly extracellular aggregation, or if NPs are being internalized and reside on the cell periphery. Similarly, there was no qualitative difference in iBMDM uptake of NPs across all conditions in the competitive uptake experiments. iBMDM presence was followed by a significant decrease in myelin signal across time. Moreover, these cells showed internalization of NPs as early as two hours post-treatment (see Figure 8, panel (h)) that would continue until at least 12 hours post-treatment. These results suggest that iBMDMs may internalize NPs regardless of MD presence.

Future Directions and Considerations

Further work is warranted to assess NP-dose-dependent uptake effects in competitive models that include myelin debris, helping to clarify how NPs affect clearance mechanisms in more representative conditions. These experiments could also integrate quantitative analyses of percent uptake to report if there are any differences in either NP or myelin debris uptake at different concentrations. Furthermore, co-culture systems incorporated SCs and iBMDMs, or in vivo studies in PNI models could provide better insight into functional consequences of NP treatment on cellular interactions and neuroregeneration. Finally, studies assessing possible interactions between electrospun CS nanofibers and CS-NPs may reveal synergistic effects between CS nanomaterials.

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